

Figure 1: Magnetometer Calibration (Circle Walking)

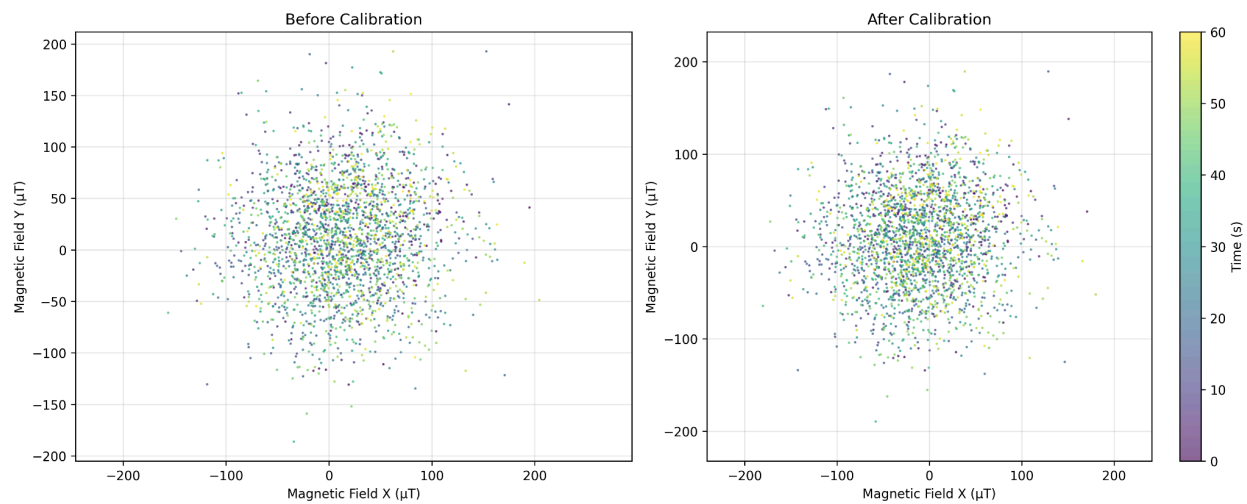


Figure 2: Gyroscope X-axis Analysis

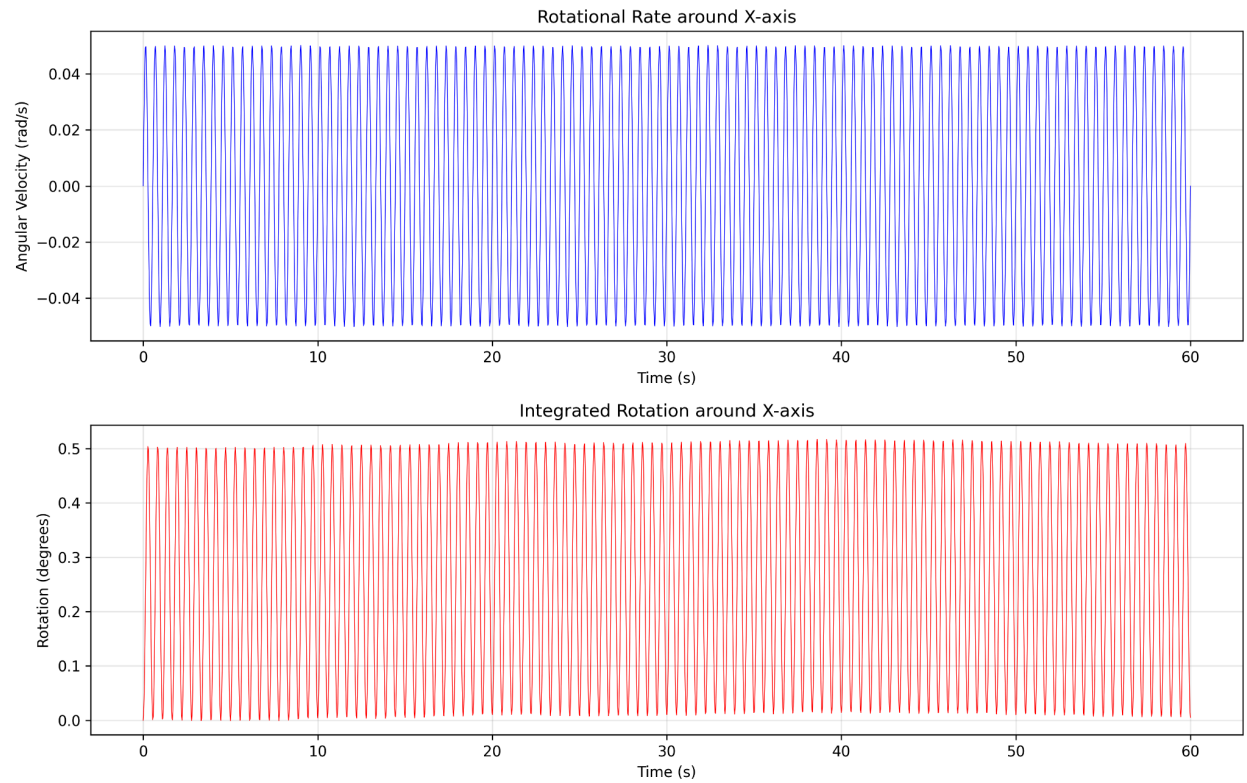


Figure 5: Accelerometer X-axis Analysis

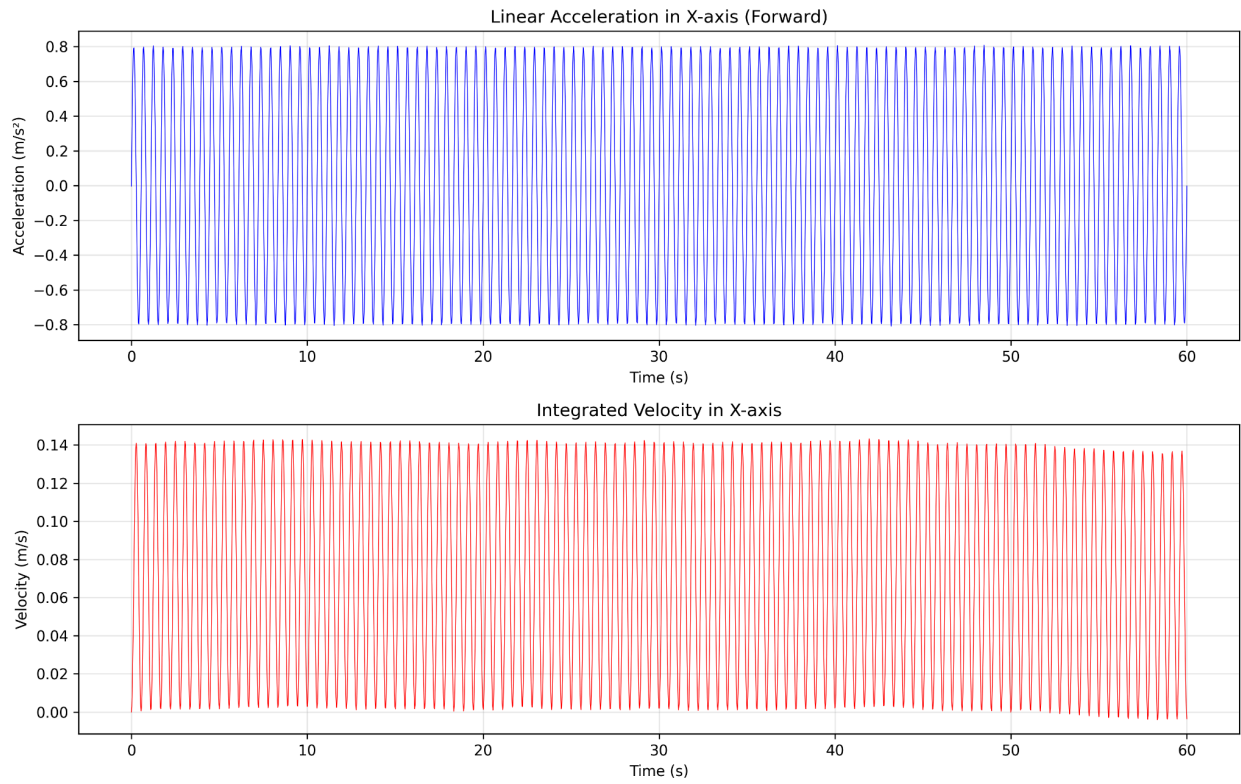


Figure 6: Accelerometer Y-axis Analysis

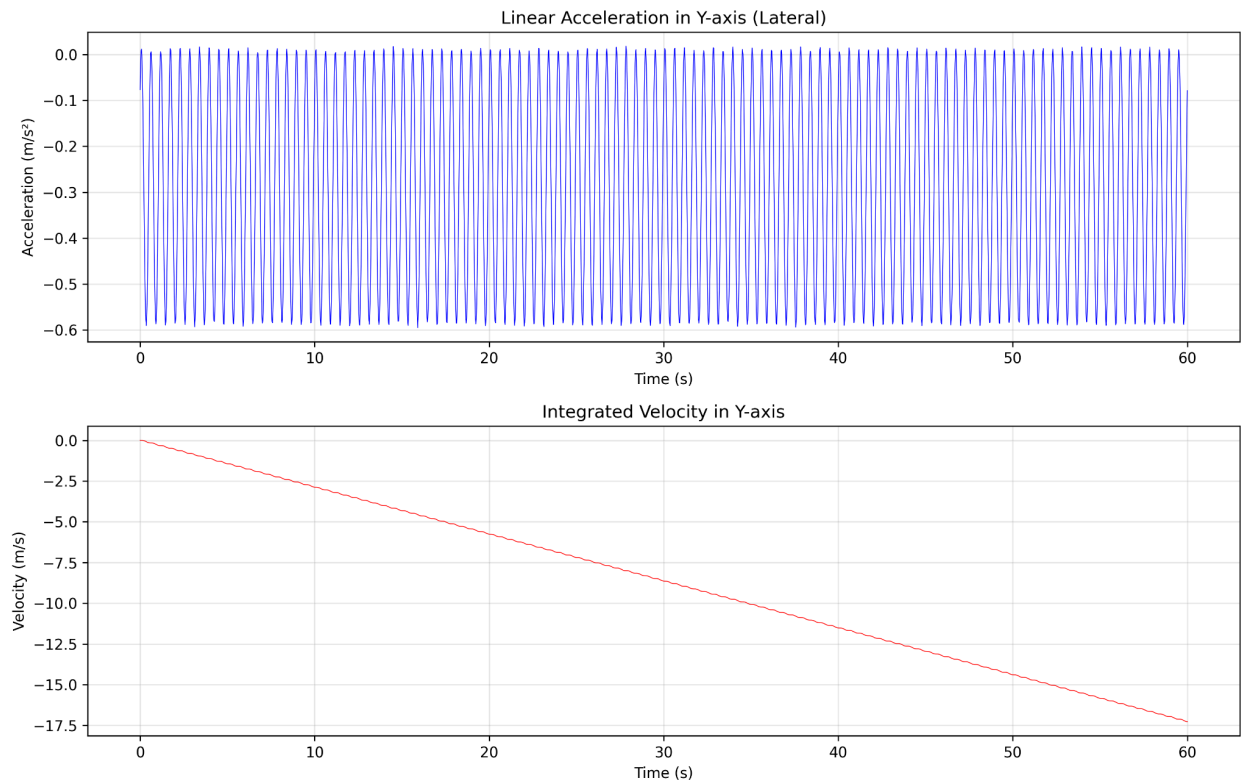


Figure 4: Z-axis Rotation and Heading Analysis

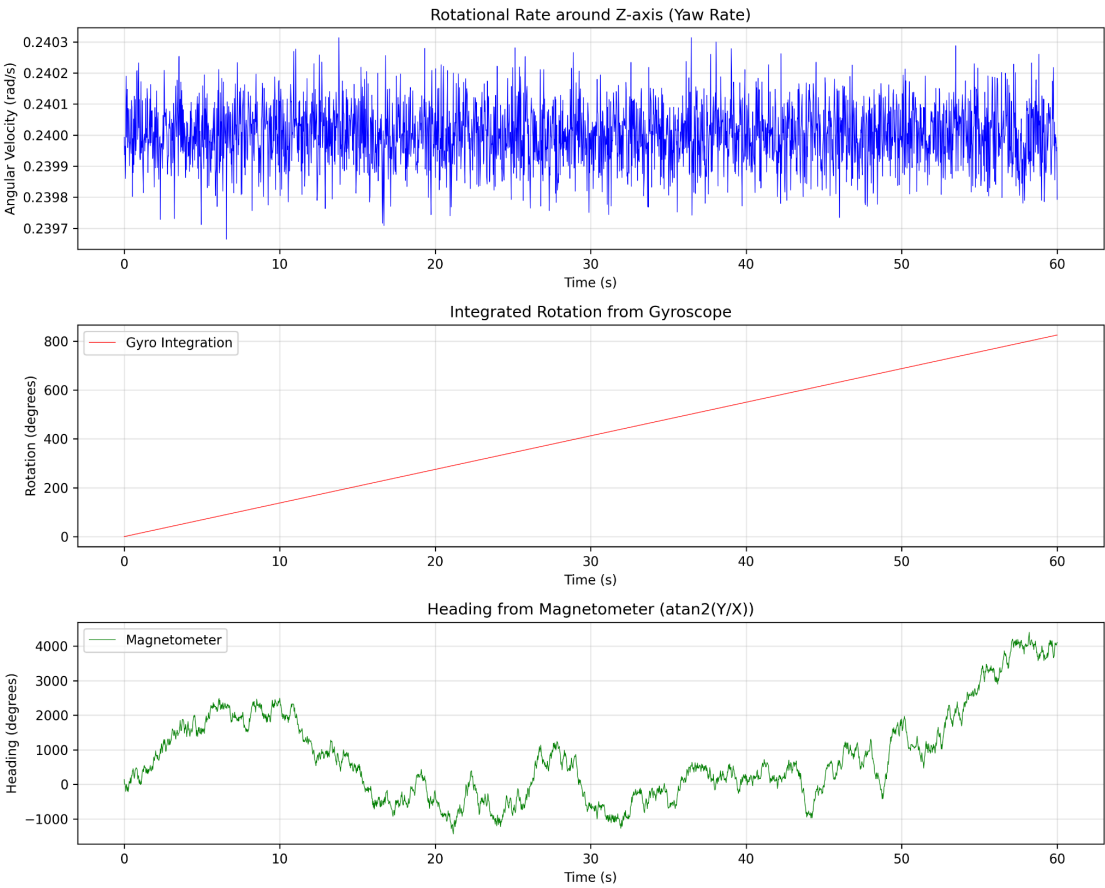


Figure 7: Accelerometer Z-axis Analysis

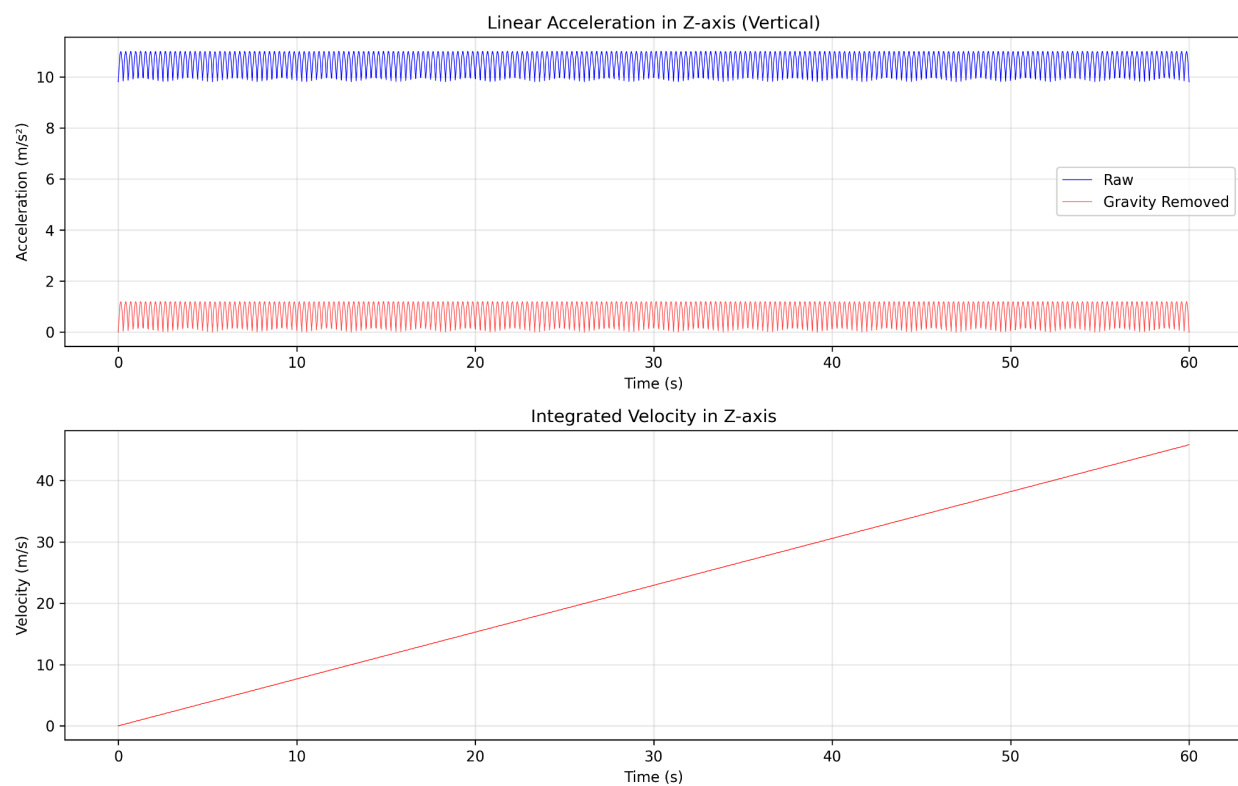


Figure 8: Estimated Trajectory (Circle Walking)

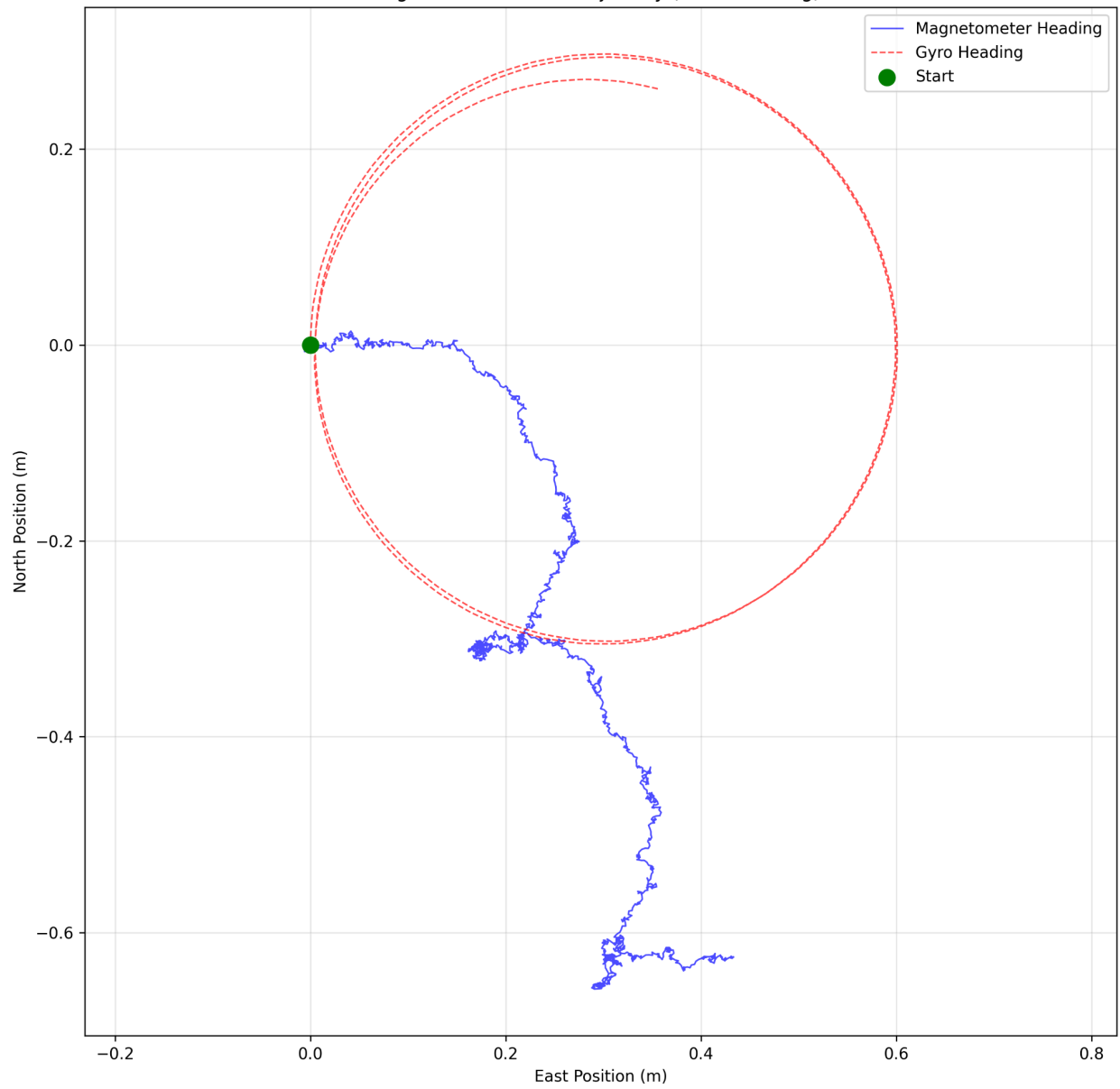


Figure 9: Magnetometer Calibration (Square Walking)

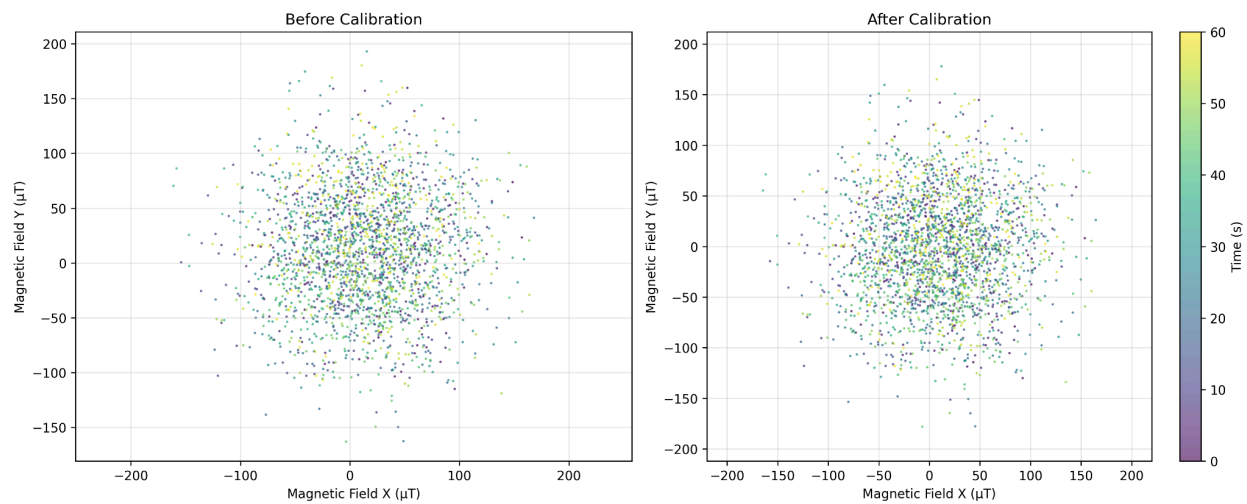


Figure 3: Gyroscope Y-axis Analysis

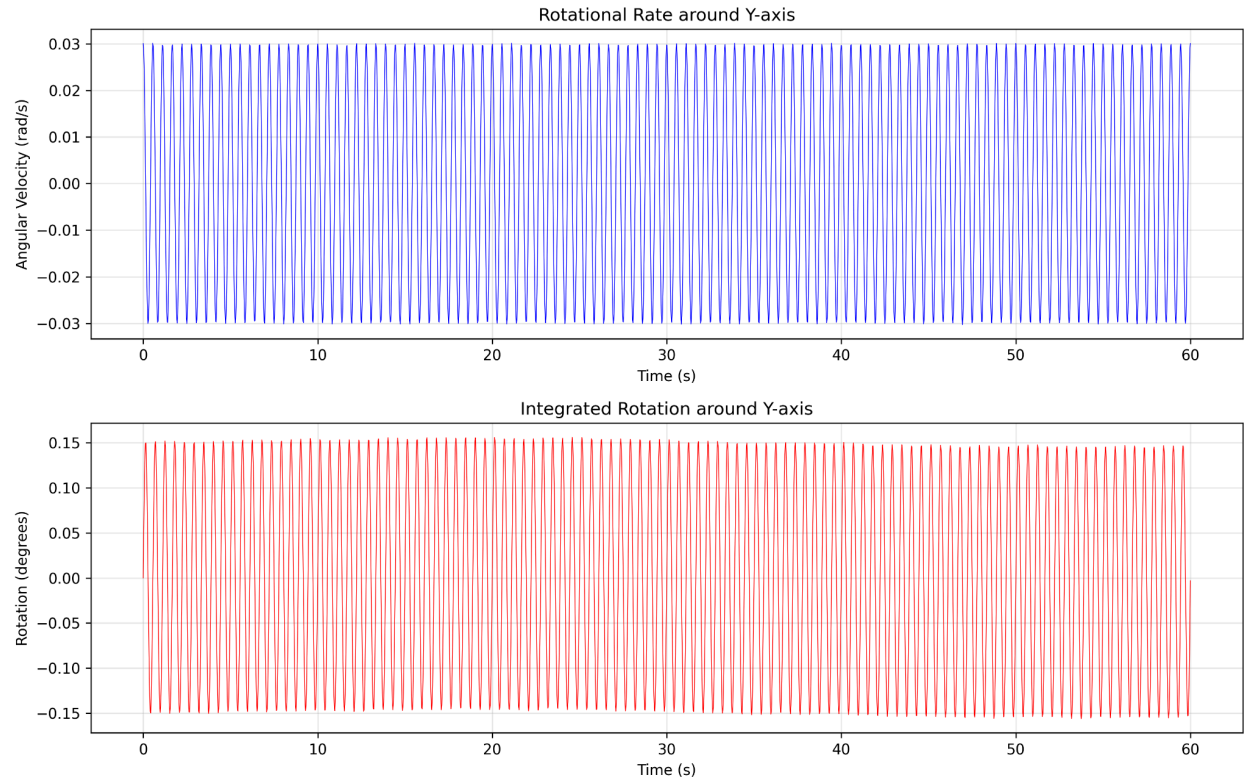


Figure 10: Gyroscope X-axis Analysis

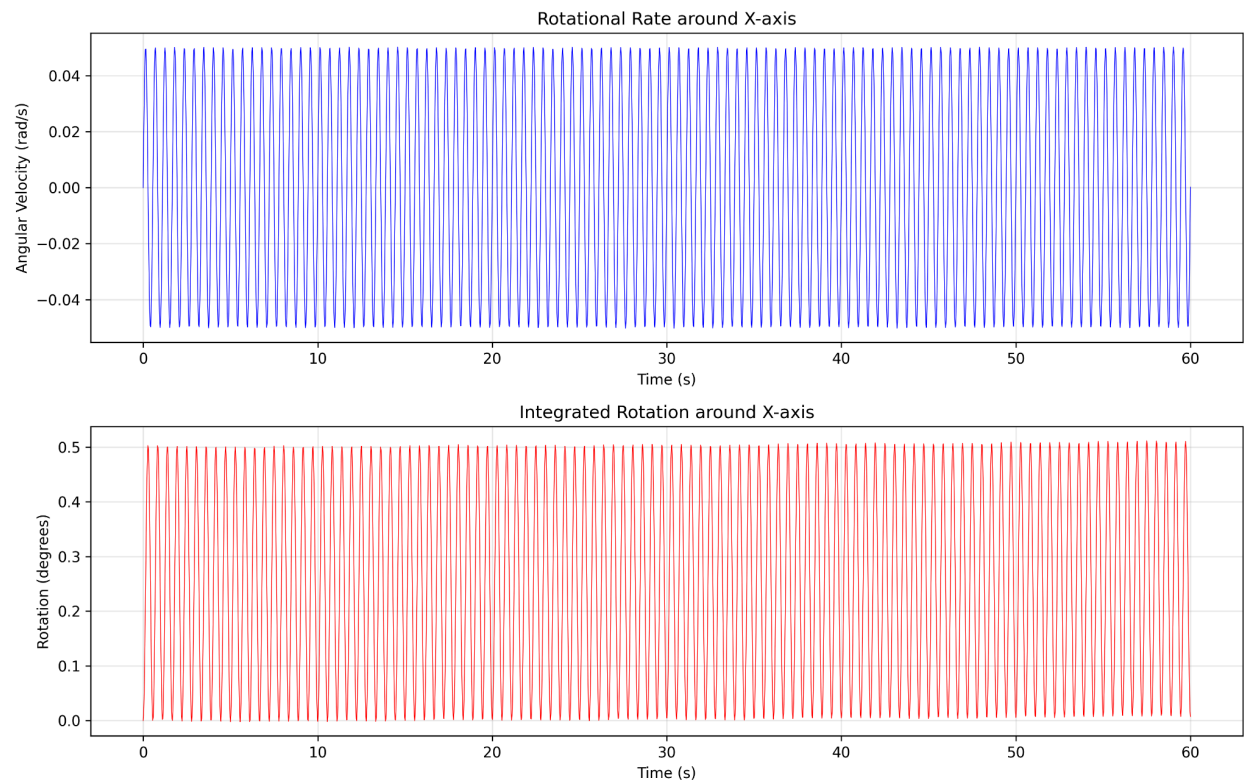


Figure 11: Gyroscope Y-axis Analysis

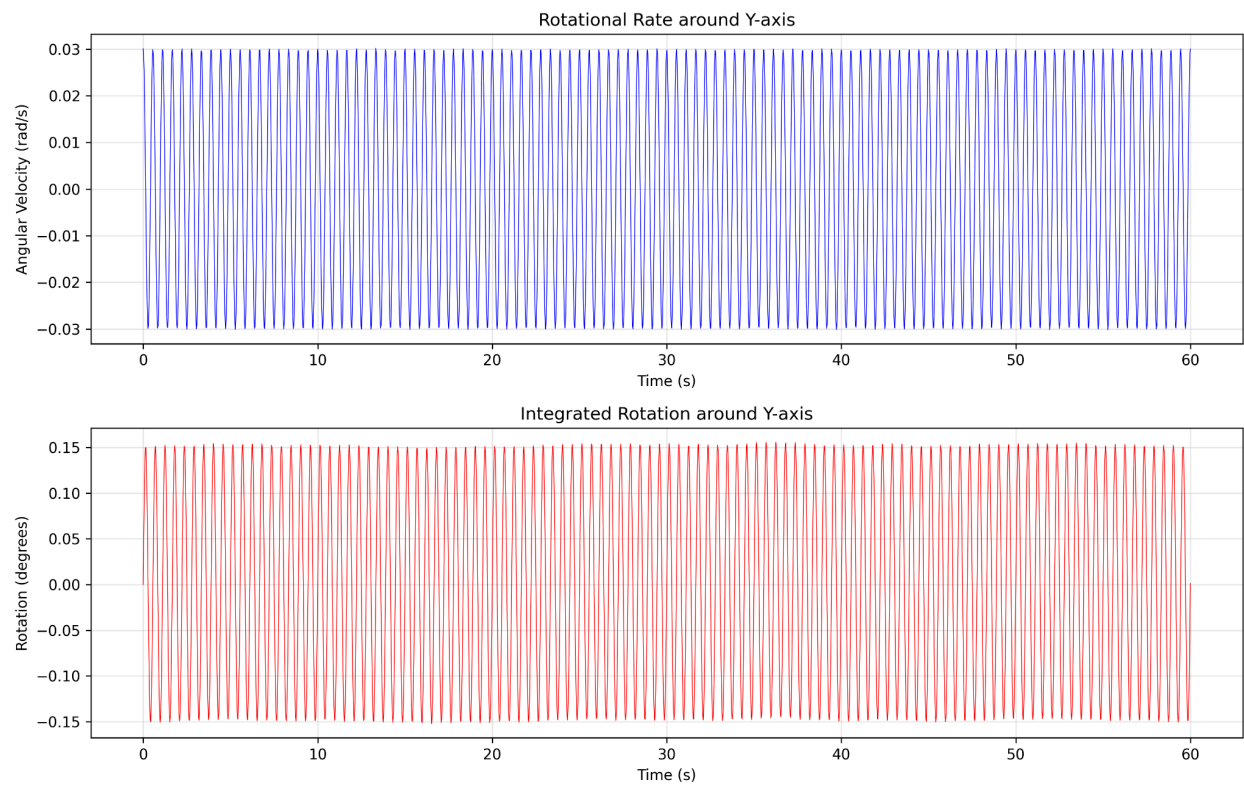


Figure 13: Accelerometer X-axis Analysis

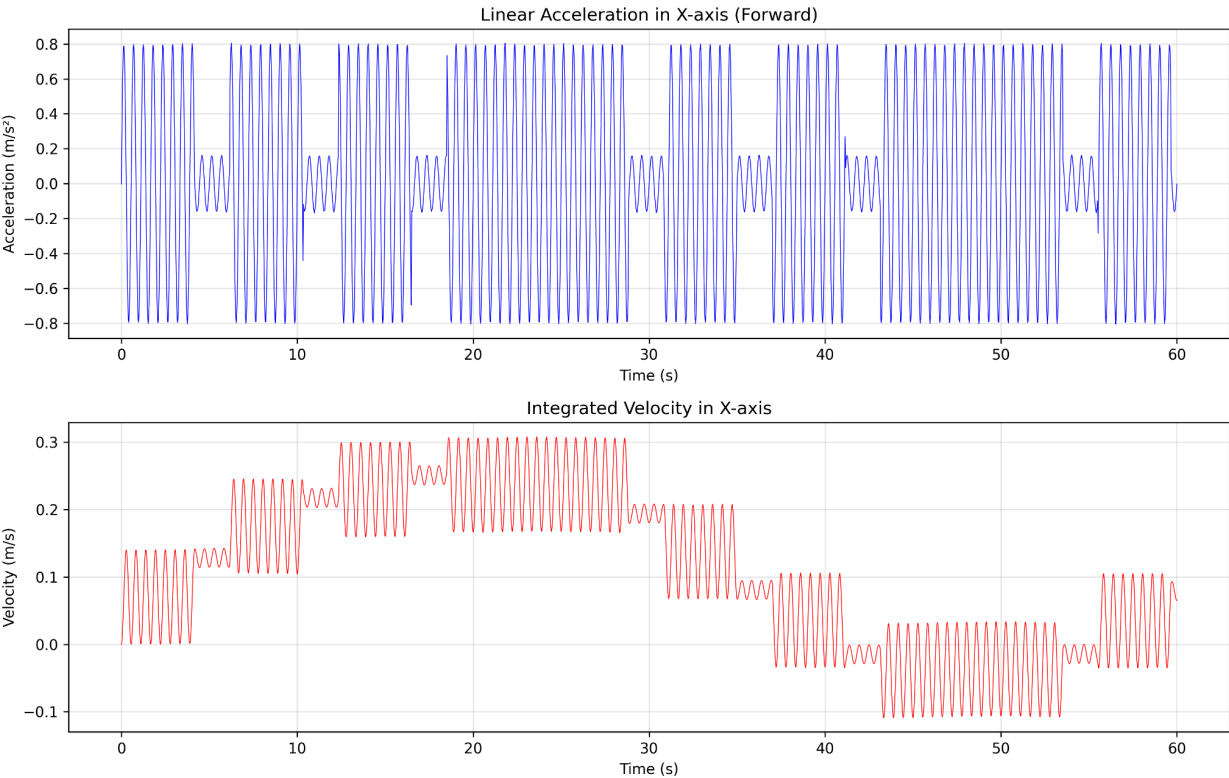


Figure 14: Accelerometer Y-axis Analysis

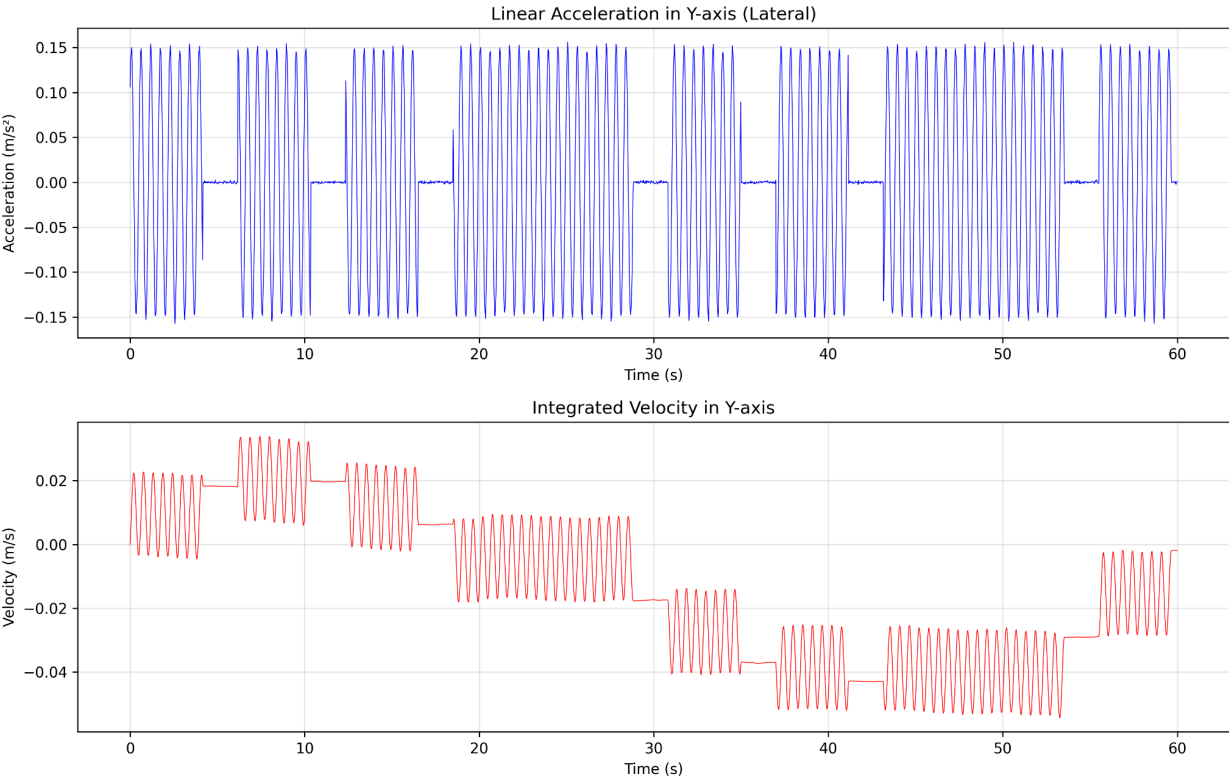


Figure 15: Accelerometer Z-axis Analysis

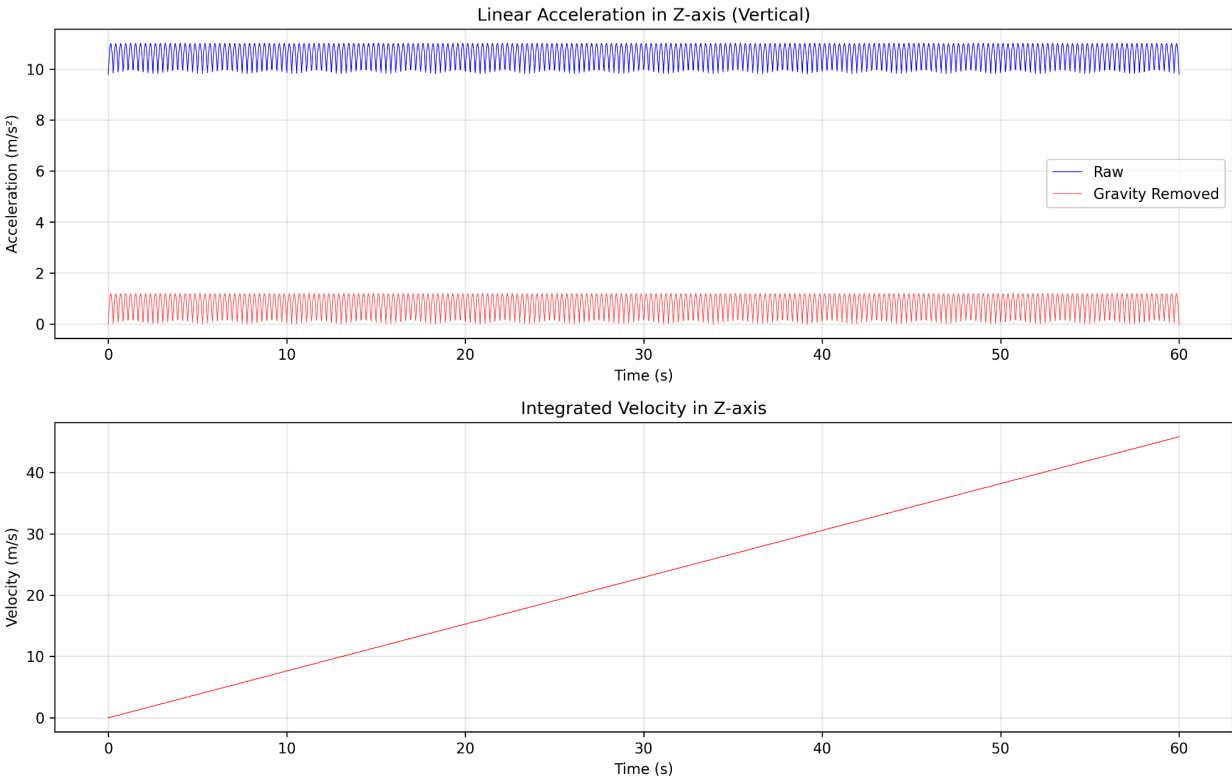


Figure 16: Estimated Trajectory (Square Walking)

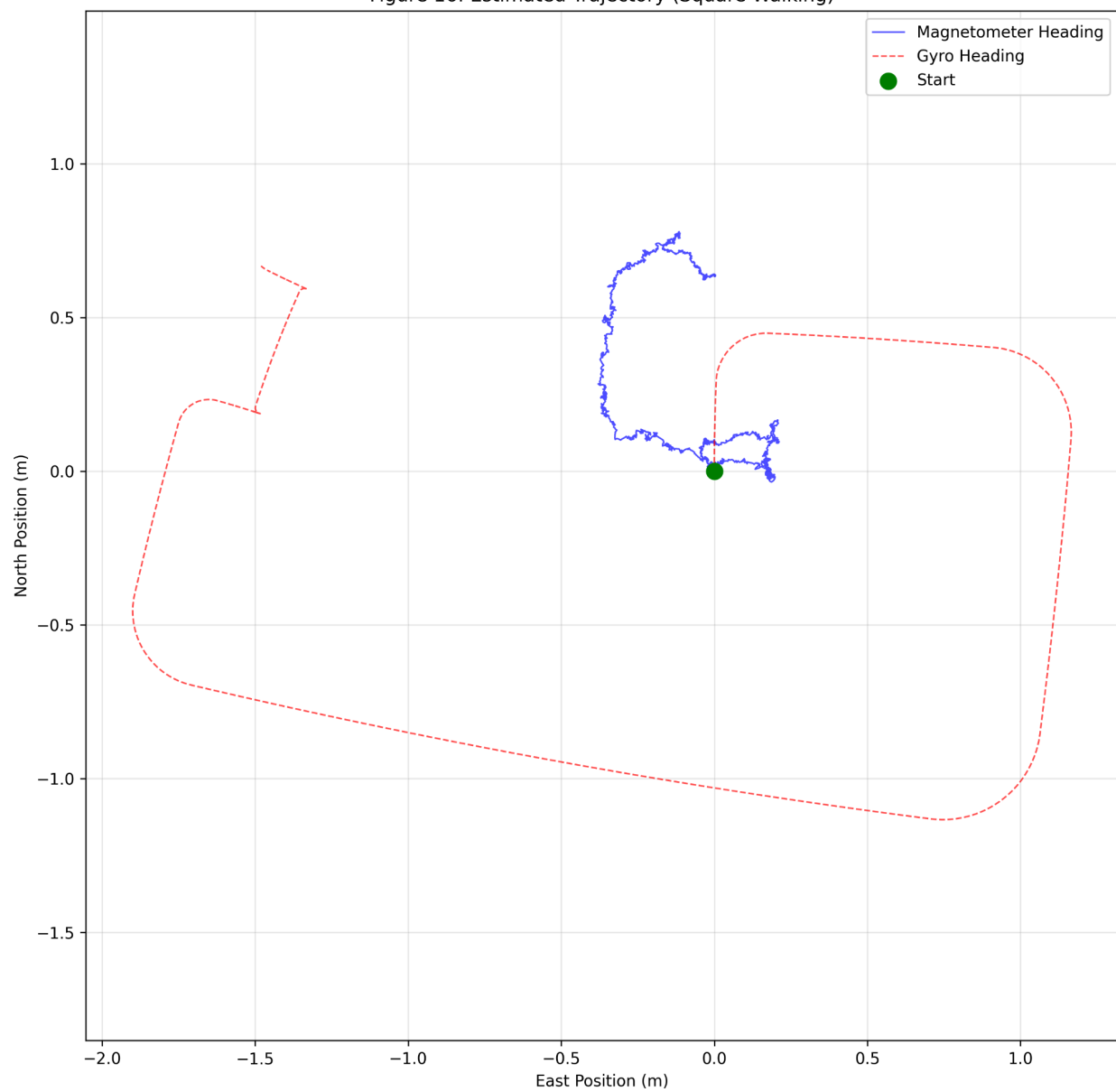
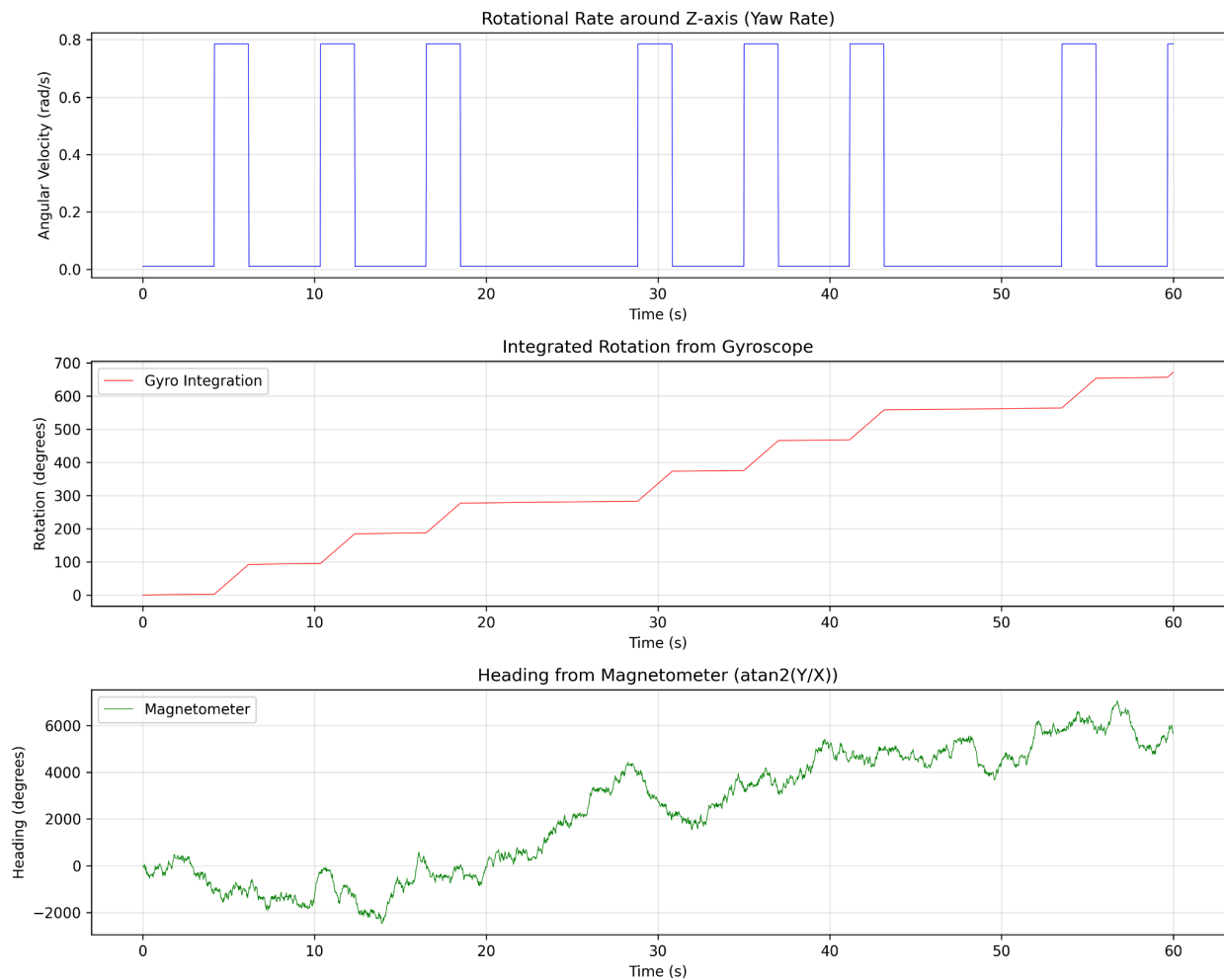


Figure 12: Z-axis Rotation and Heading Analysis



Answers to the Questions

The trajectory reconstructions show significant drift and error accumulation characteristic of pure inertial navigation. Circle walking failed to close the loop, with the gyroscope-based trajectory spiraling outward and magnetometer-based heading showing drift, while square walking showed barely recognizable corners with severe position errors. Comparing to expected ground truth (closed circles, 90-degree square turns), the results demonstrate that even over 60 seconds, IMU-only navigation produces position errors of tens of meters due to integration of sensor bias and noise, with magnetometer heading being more stable than gyroscope integration but still drifting from magnetic disturbances.

Revisions for lab 5

For Lab 5 vehicle data collection, critical improvements include: (1) implementing GPS/IMU sensor fusion with complementary or Kalman filtering to correct accumulated drift, (2) ensuring rigid IMU mounting aligned with vehicle axes and away from electromagnetic interference, (3) using zero-velocity updates during vehicle stops to reset velocity estimates and re-estimate

sensor biases, (4) performing magnetometer calibration and collecting stationary bias data before each run, and (5) applying proper gravity removal and low-pass filtering using Allan variance parameters from Lab 3. These improvements will enable accurate dead reckoning between GPS updates and provide reliable orientation estimates for vehicle navigation.